

STATSEI14 · PRACTICAL TUTORIAL

Deep Learning Meets ETAS

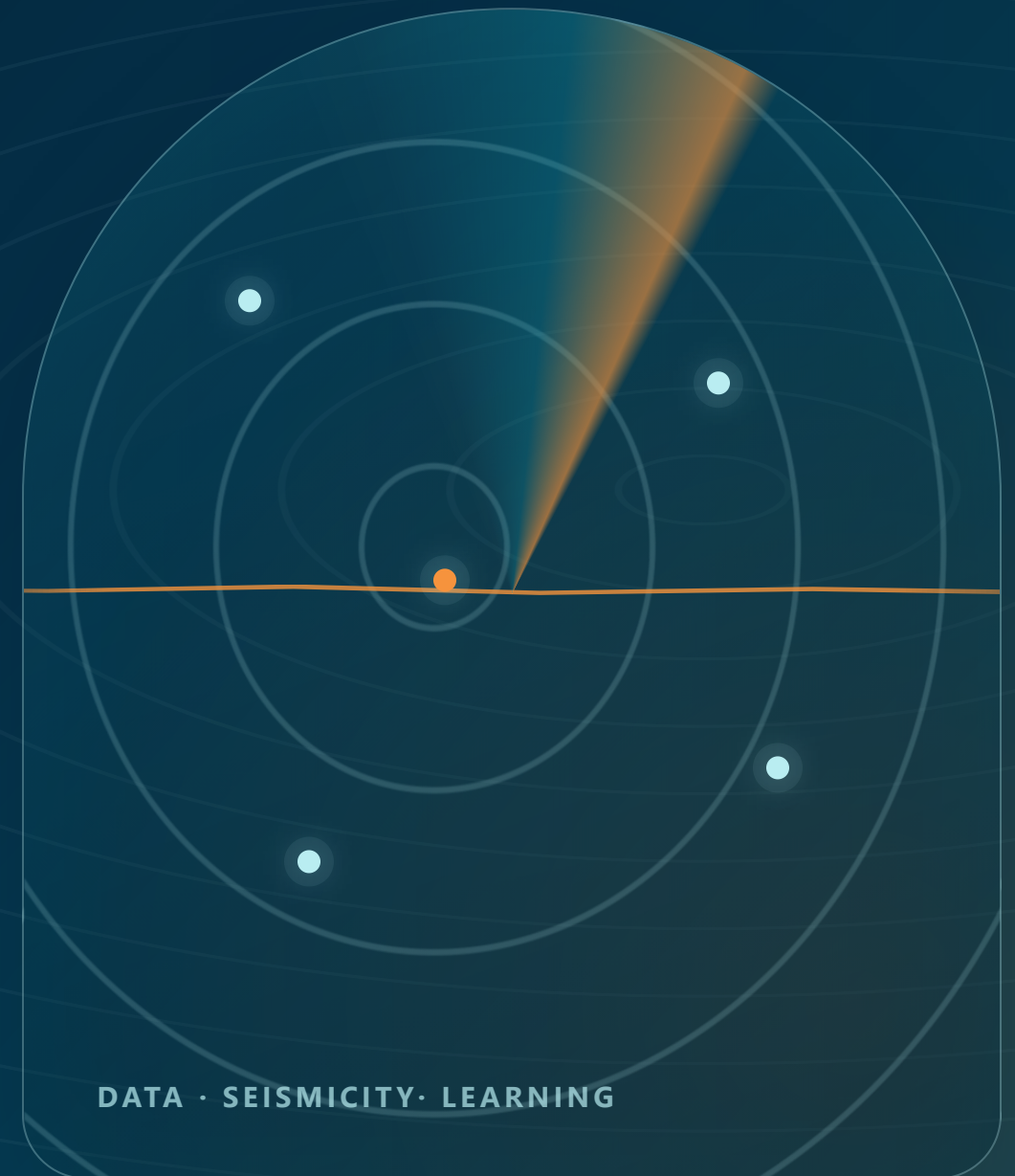
Practical Hybrid Workflows for Seismic Forecasting

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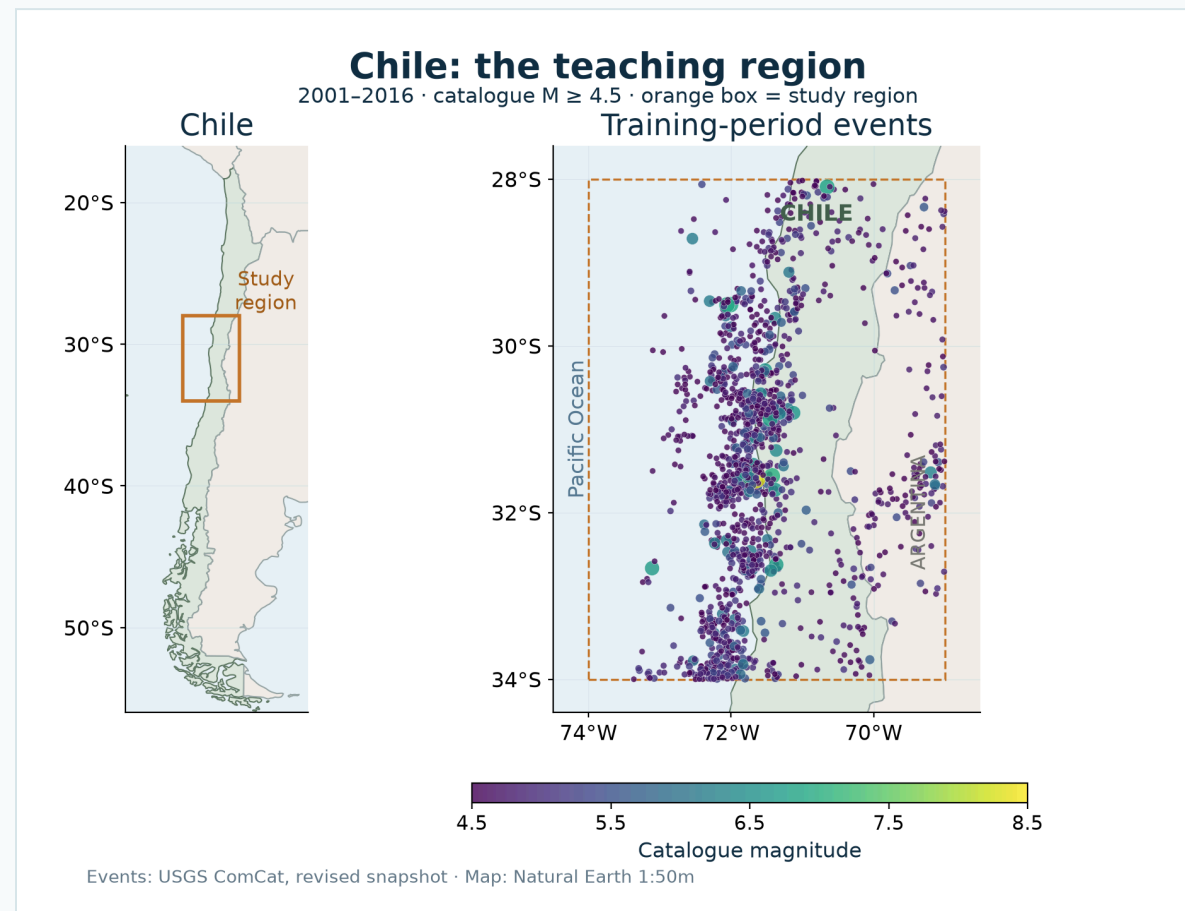
13 October 2026 · Chile

Seismic learning tasks · Research cases · Guided modelling and discussion



DATA · SEISMICITY · LEARNING

What can we learn from seismicity?



Signals reveal recorded events.

Catalogues describe event histories.

Models turn histories into questions about what may follow.

What information do we represent, and what do we ask a model to learn?

Map: frozen USGS training catalogue and Natural Earth. Illustrates today's regional setting.

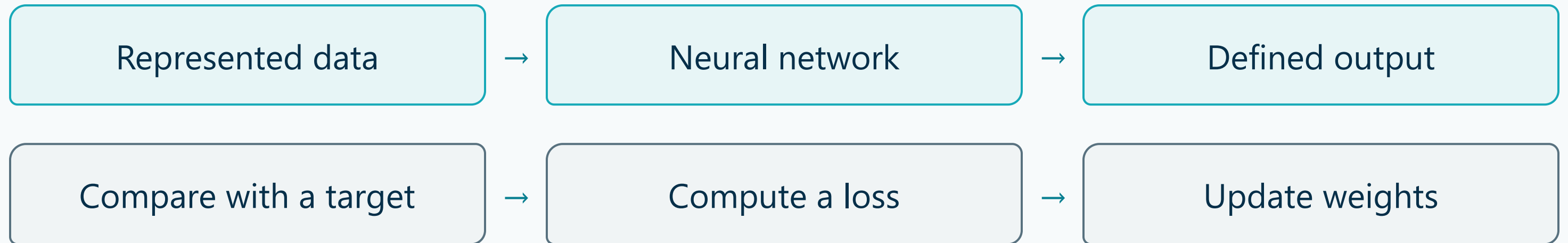
Deep learning in seismology: inputs and targets

Example	Input	Learning task
ConvNetQuake · 2018	Recorded waveforms	Detect events and classify their source region
RECAST · 2023	Past event times and magnitudes	Estimate the next-event time distribution
Delgado et al. · 2025	Features of catalogued events	Classify foreshock, mainshock and aftershock labels
Farfán et al. · 2025	Sequences of daily spatial grids	Forecast gridded seismic activity

Today’s practical: 30 days of history and a seven-day occurrence probability.

Perol et al. · Dascher-Cousineau et al. · Delgado et al. · Farfán et al.

A network learns a mapping from examples

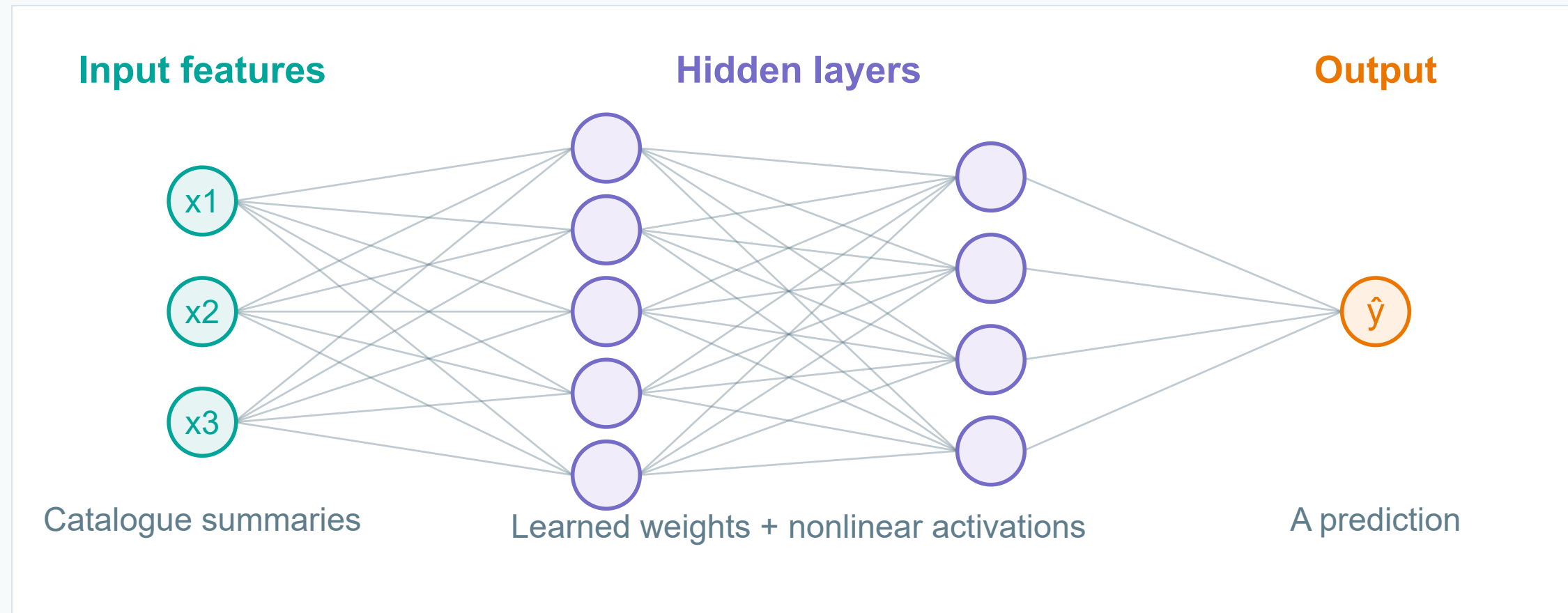


Architecture determines how information is combined.

Training adjusts weights to reduce the loss.

The output and the loss must match the scientific question.

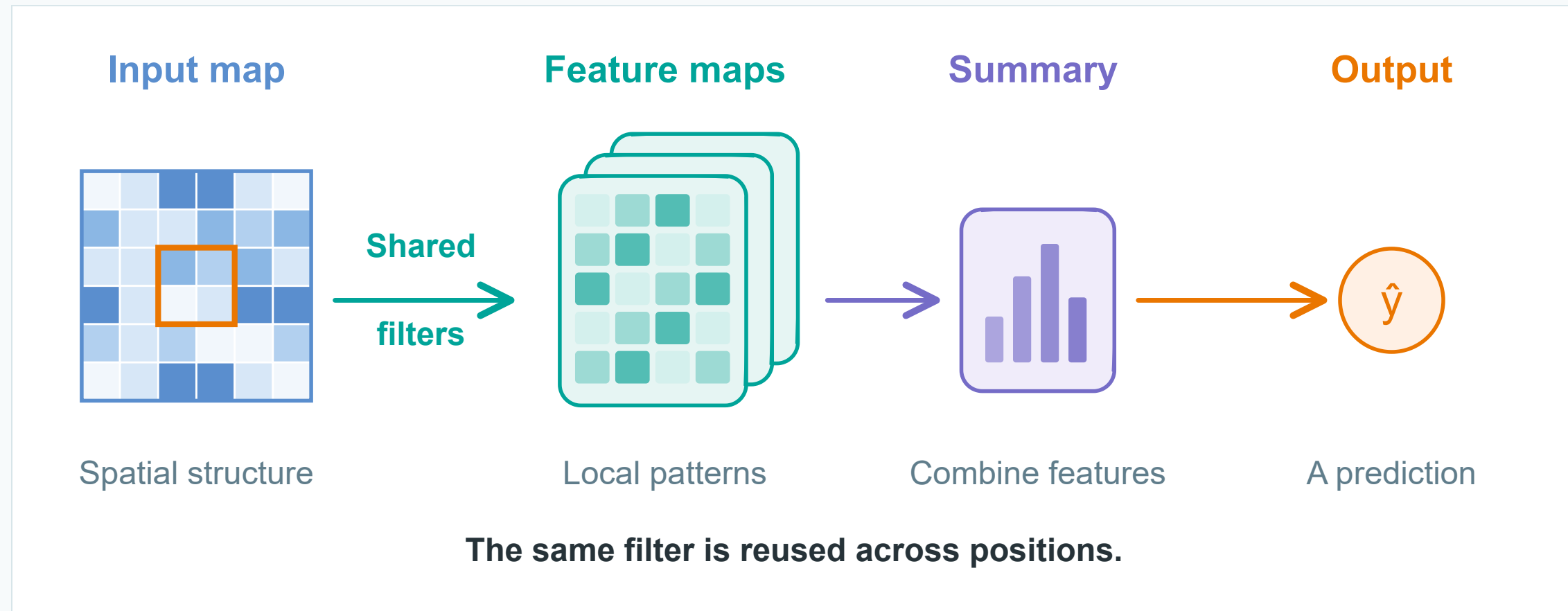
MLP: combine measured features



A multilayer perceptron learns nonlinear combinations of a feature vector.

Temporal laboratory: the MLP uses 90 values at fixed lag positions. Diagram adapted from the author's DL course, Unit 2.

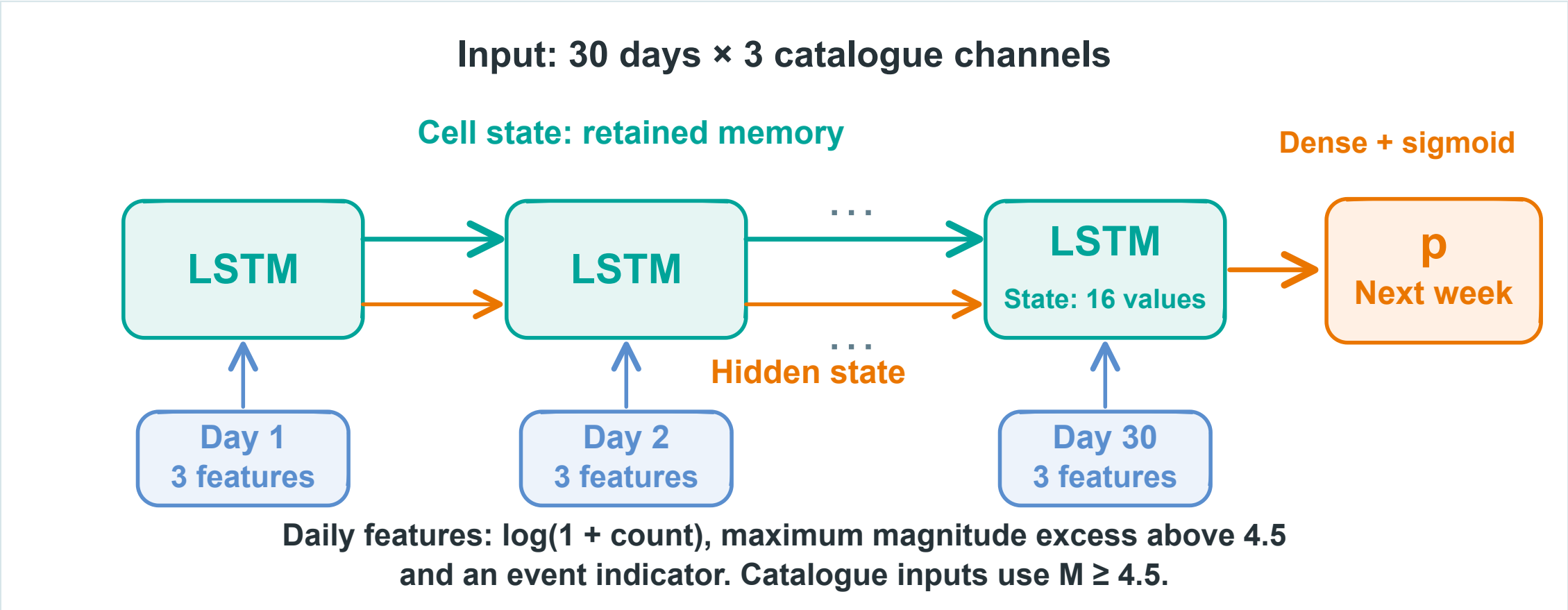
CNN: learn local patterns with shared filters



The same filter examines different neighborhoods.

Temporal laboratory: CNN1D shares filters across days. Diagram adapted from the author's DL course, Unit 3.

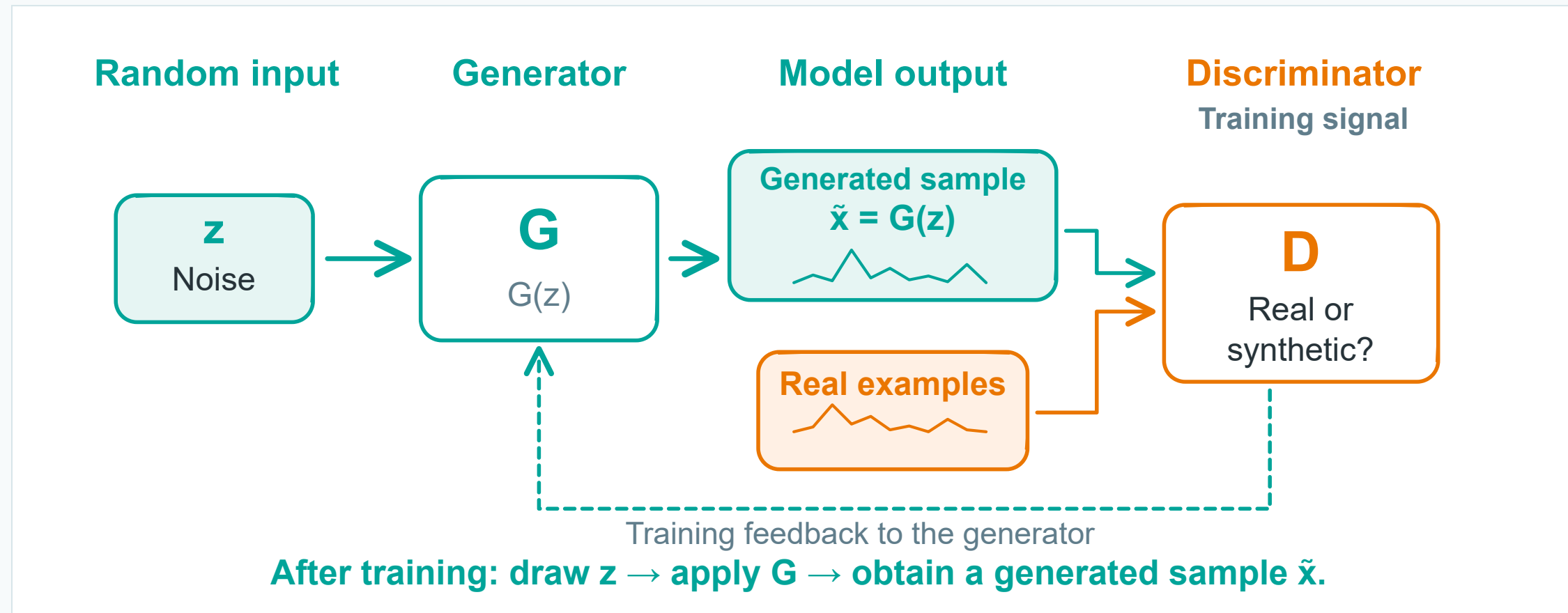
LSTM: carry information through a sequence



Today's example: 30×3 input → state (16) → Dense + sigmoid → weekly probability

Output: probability of at least one catalogue $M \geq 5$ event in the next seven days.
Adapted from the author's DL course, Unit 4.

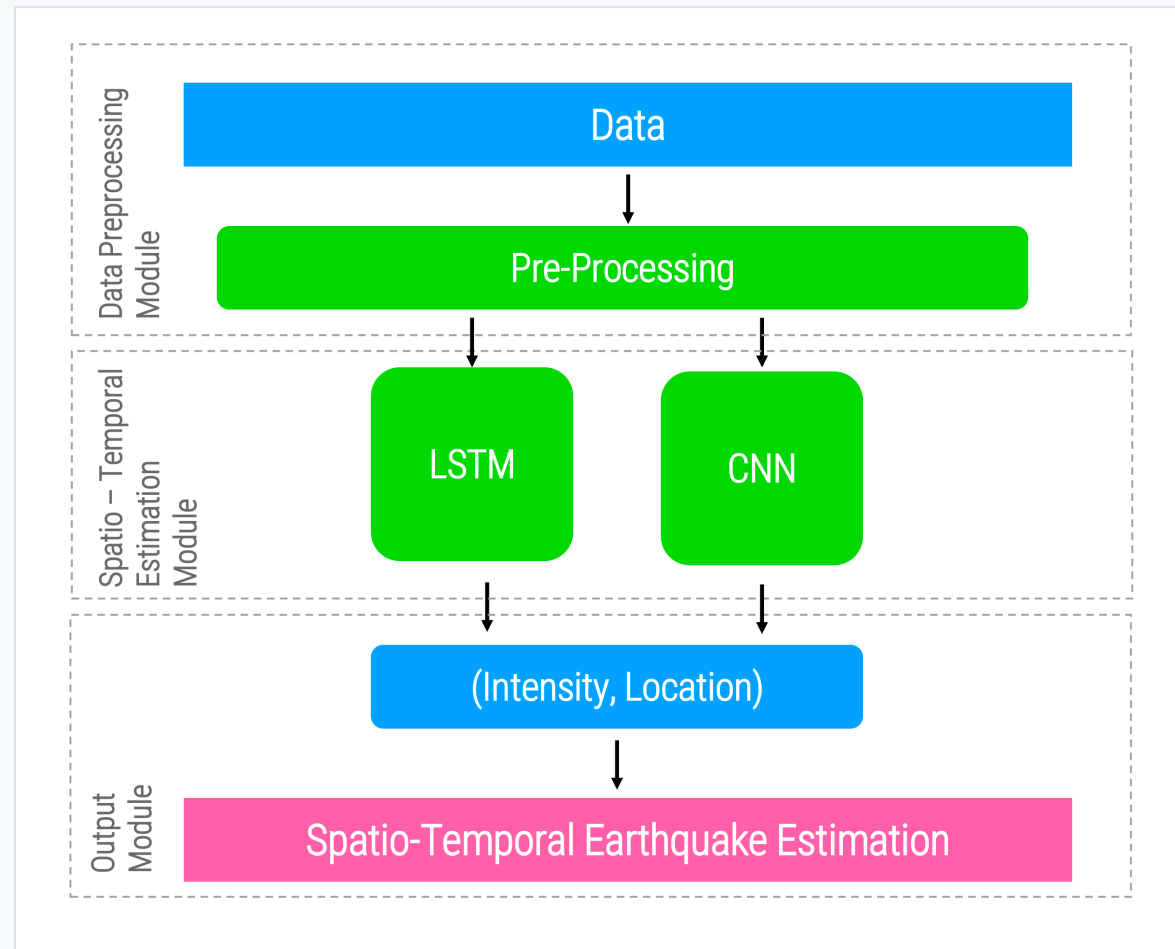
Generative learning: produce new samples



The output of G is a new data sample, such as a synthetic sequence.

Later case: mainshock conditions + noise $\rightarrow$ a possible aftershock sequence.
GAN schematic adapted from the author's DL course, Unit 5.

Our first case: learning from an ETAS representation



Scientific question

Can neural networks learn the evolution of a statistical representation of seismic activity?

Two views

A daily time series and a sequence of spatial maps.

Nicolis, Plaza & Salas · Spatial Statistics 42 (2021), 100442 · CSN Chile, 2000–2017

ETAS supplies a statistical description of activity

$$\lambda(t \mid \mathcal{H}_t) = \mu + \sum_{t_i < t} \text{contribution from event } i$$

Background + magnitude-dependent contributions that decay with time

Spatial ETAS also describes how contributions vary over space.

This conditional occurrence rate becomes a representation for learning.

Conceptual sketch. Process intensity is different from magnitude and shaking intensity.

The temporal task: tomorrow's maximum intensity



Input and target: summaries of the ETAS-derived intensity.

Learning question: how much of its evolution can a temporal network capture?

Specify what is being predicted before interpreting a fit statistic.

Published case, Nicolis, Plaza & Salas (2021). The target is a model-derived quantity.

The spatial task: the macrozone of the maximum

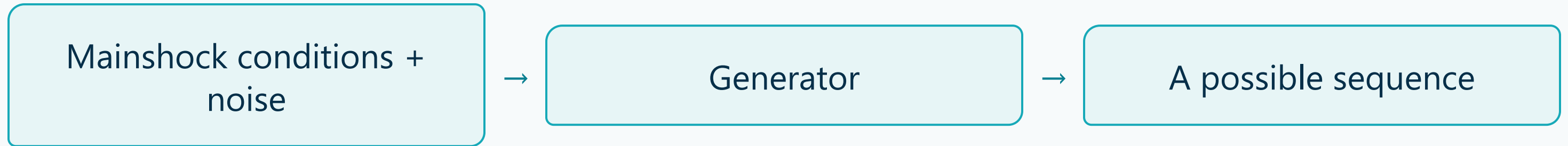


Which macrozone contains tomorrow's maximum estimated intensity?

The representation determines which information the network can use.

Published case, Nicolis, Plaza & Salas (2021). These probabilities concern six macrozones.

With Isidora: can we generate aftershock sequences?



Thesis question

Can conditional generative models reproduce the temporal pattern of aftershock sequences?

Model families explored

Conditional GAN · WGAN-GP · attention-based generator

A richer output requires us to decide what a sequence actually represents.

Isidora Jara Muñoz (2025), thesis · Subsequent manuscript by Plaza-Vega, Jara, Nicolis & Salinas (2026), under review

One sequence mixes absence and magnitude

Representation in the manuscript

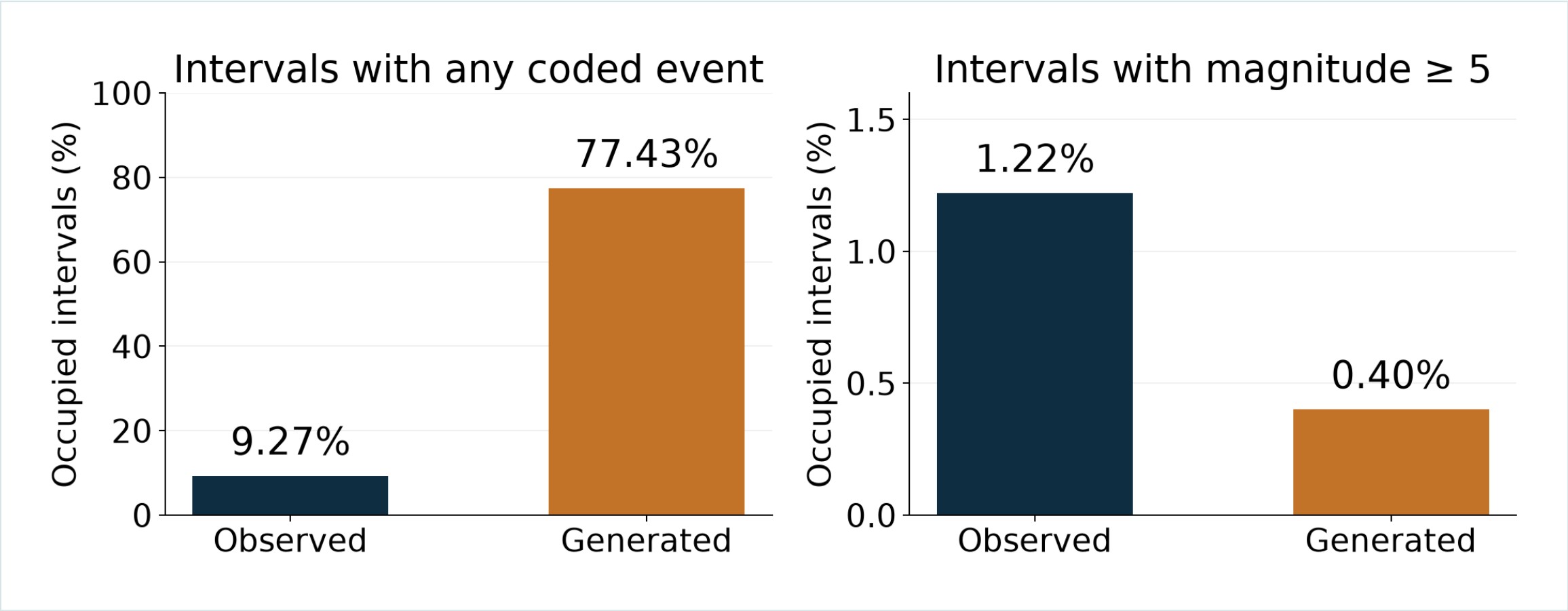
14 days → 224 intervals of 90 minutes → maximum magnitude per occupied interval



Schematic values. Gray: absence marker. Orange: maximum magnitude in an occupied interval.

“No recorded event” and “event magnitude” are different outcomes.

Generated sequences distort event occupancy



Assess which properties of the observed sequences are reproduced.

2026 manuscript under review · Occupied intervals, not total events · Source-case results

The article revisits what the model should learn

Empirical reformulation

Aggregate targets: occurrence, exceedance and count summaries.

The classification branch uses tree models.

A future neural formulation

Occurrence probability, followed by magnitude conditional on occurrence.

Evaluate both parts and their joint behavior.

Today we isolate occurrence in a fixed future interval as the learning target.

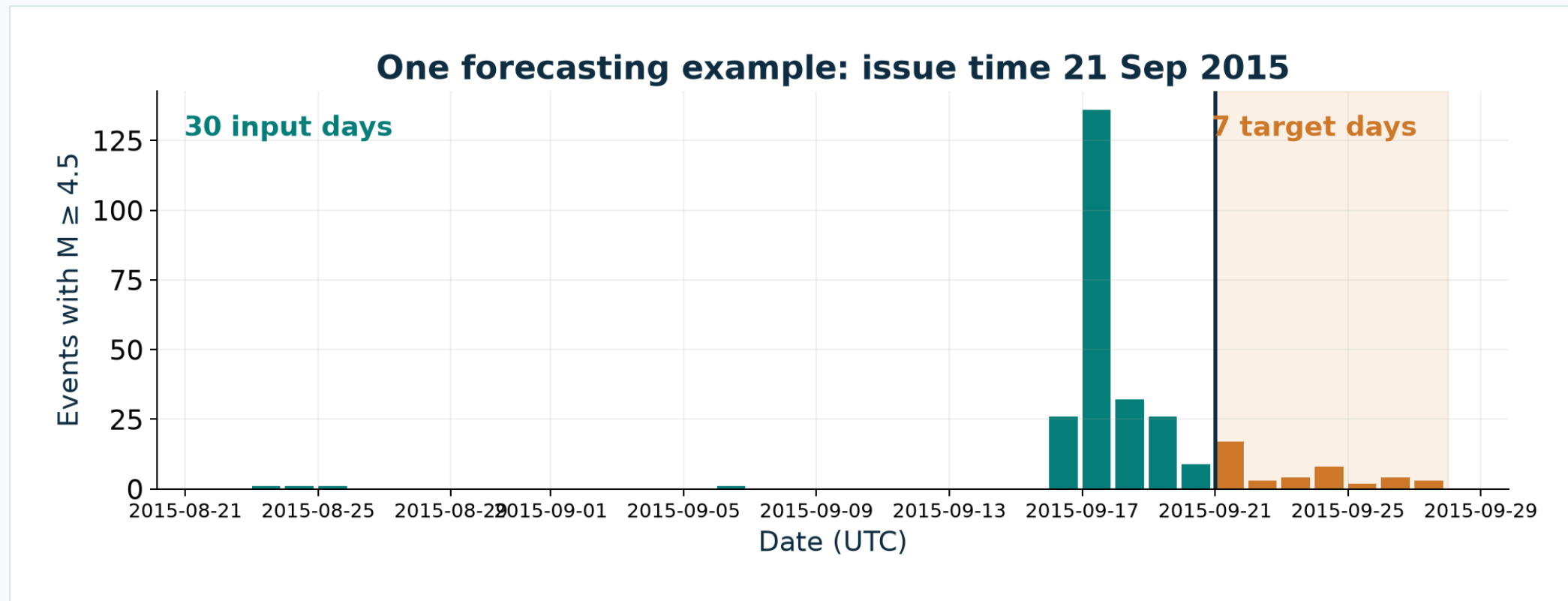
ETAS-informed generation remains future work. The two-part neural model is a proposal.

The practical puts the same decisions in your hands

Case	Representation	Output
Spatial Statistics	ETAS time series and maps	Intensity maximum and its macrozone
Isidora / manuscript	Mainshock conditions; binned sequences	Conditional sequences; aggregate targets
Today	Daily catalogue channels	Weekly occurrence probability

Live: LSTM and Transformer. Laboratory extensions: MLP, CNN1D, maps and graphs.

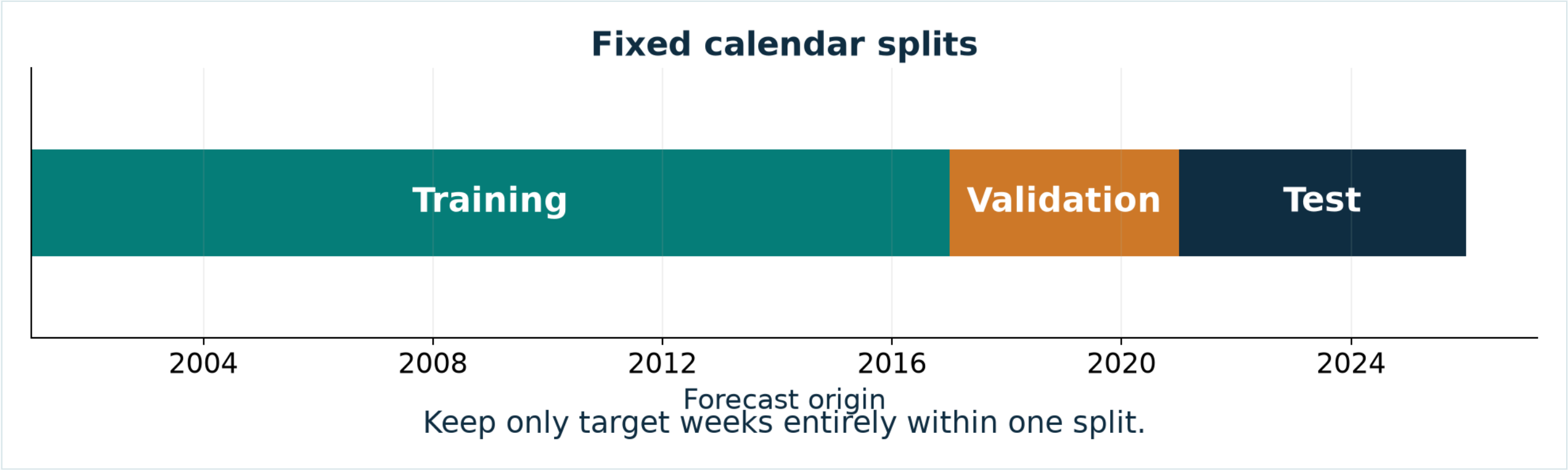
Today's target: at least one event next week



30 past days → probability of at least one catalogue $M \geq 5$ event in seven days.

One input: 30 days × 3 channels · Region: 28–34°S, 69–74°W · Input $M \geq 4.5$ · Monday origins

Evaluate using the temporal order of the problem



Today we compare validation probabilities using Brier score, log loss and calibration.

Train 2001–2016 · Validation 2017–2020 · Test 2021–2025 was inspected previously and is excluded from this comparison.

Experiment: how does the same history enter each model?

30 days × 3 catalogue channels → LSTM → weekly probability

The same history → Transformer → the same target

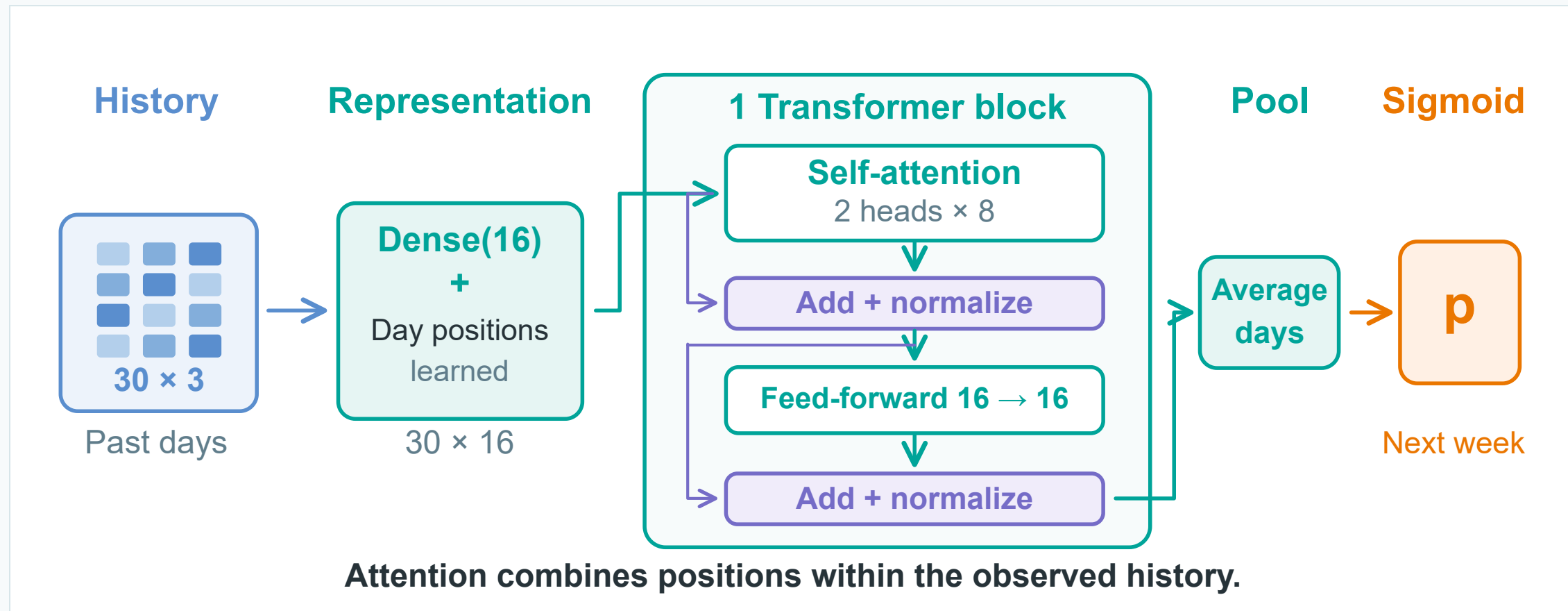
Build one example: identify the input window, its origin and its label.

Train and evaluate: read `compile`, `fit`, `predict` and compare probability errors.

What changes when different architectures process the same information?

[Temporal tutorial and notebook](#) · [Seed 7](#) · [MLP and CNN1D for the laboratory](#) · [Activity-score experiment as an extension](#)

A Transformer combines the same history differently



Same history and target. A different way to combine information across days.

Current temporal tutorial · 2,257 trainable parameters versus 1,297 for LSTM · One fixed, small configuration

Validation performance for the weekly occurrence task

Model	Validation Brier	Validation log loss
Training prevalence	0.187690	0.562855
Logistic regression	0.203664	0.614405
LSTM	0.181521	0.547512
Transformer	0.187901	0.562477

Lower is better. Compare these scores with the calibration plot and its sample counts.

These are development results from one region and one validation period.

One training run per neural model, seed 7 · 208 validation weeks

What can this result tell us about the problem?

Reading the validation results

Compare each model with training prevalence.

Read Brier, log loss and calibration together.

Questions for the next experiment

Which representation and target preserve useful structure?

What information is available in the catalogue?

Would more data, capacity or a different training budget change the result?

Useful deep learning depends on how we represent the phenomenon and formulate the task.

Validation also controls early stopping. One seed does not measure initialization sensitivity.

What did we learn, and what would we test next?

1. What information did our daily channels preserve or discard?
2. What did the LSTM and Transformer comparison establish?
3. Which extension would you test next: maps, graphs or an activity score?

Deep learning offers flexible tools. Their scientific value depends on the question and the evidence.

Homework: [guided reading](#) or a [controlled extension](#) · Build on today's LSTM and Transformer comparison

Appendix · the LSTM steps

```
1 inputs = keras.Input(shape=(30, 3))
2 state = keras.layers.LSTM(16)(inputs)
3 probability = keras.layers.Dense(1, activation="sigmoid")(state)
4 model_lstm = keras.Model(inputs, probability)
5 model_lstm.compile(optimizer="adam", loss="binary_crossentropy")
6 history_lstm = model_lstm.fit(
7     X_train, y_train, validation_data=(X_val, y_val),
8     epochs=20, batch_size=64, callbacks=[stopping_lstm], shuffle=False)
9 p_lstm = model_lstm.predict(X_val).ravel()
10 brier = np.mean((p_lstm - y_val) ** 2)
```

Define the model · compile · fit · predict · evaluate probabilities

Appendix · quantities and scientific meaning

Quantity	Question answered
Process intensity	At what conditional rate do events occur?
Magnitude	How large is an event?
Shaking intensity	What motion or effects occur at a location?
Risk	What consequences may occur given exposure and vulnerability?

Today's output is a probability of a defined occurrence event.

Appendix · ETAS-inspired feature experiment

$$r_t(s) = \sum_{t-L \leq t_i < s} 10^{\alpha(M_i - M_0)} \left(1 + \frac{s - t_i}{c}\right)^{-p}, \quad s \leq t.$$

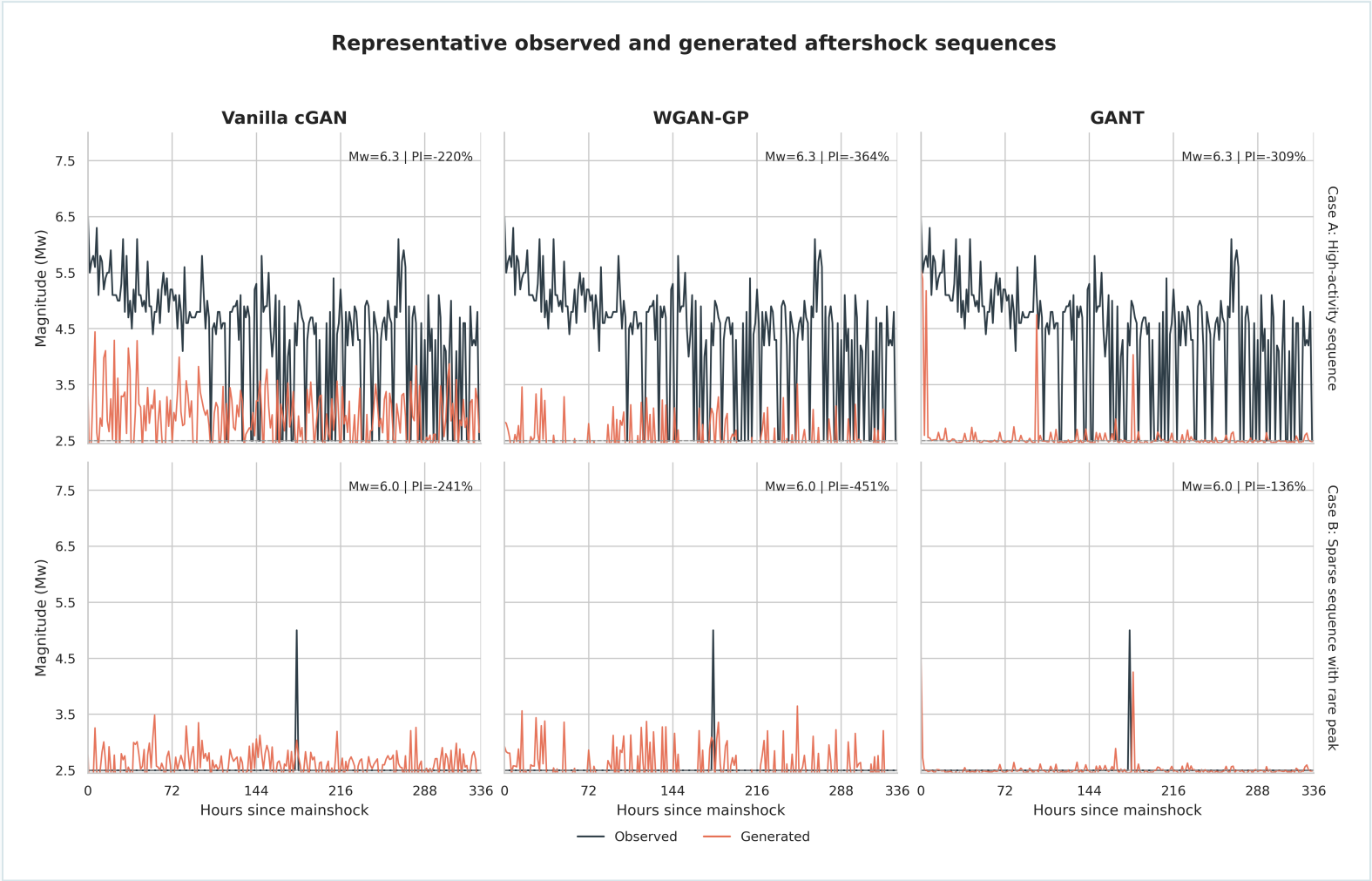
Illustrative: $\alpha=1$, $c=1$ day, $p=1.1$, $L=30$ days.

Controlled comparison: both LSTMs use the same available history.

Representation: $\log_{10}(\text{score})$ is an additional daily channel.

Optional representation experiment. The illustrative index is not a fitted ETAS rate.

Appendix · original generative examples



2026 manuscript under review · Original PI annotations use a mean-reference denominator · Illustrative cases

Appendix · evidence and availability

Check	Meaning
Chronological, purged targets	Future target intervals do not cross partitions.
Training-only normalization	Later inputs do not set scaling parameters.
Frozen revised catalogue	Data bytes and provenance are reproducible.
Historical catalogue vintages	Not reconstructed.

Reproducibility and historical information availability are separate requirements.

Appendix · sources and further reading

- **Seismic applications:** [ConvNetQuake](#) · [RECAST](#).
- **Our published case:** [Nicolis, Plaza & Salas \(2021\)](#).
- **Generative case:** [Jara Muñoz \(2025\)](#), thesis; [Plaza-Vega, Jara, Nicolis & Salinas \(2026\)](#), manuscript under review.
- **Attention context and further reading:** [Delgado et al. \(2025\)](#) · [Farfán et al. \(2025\)](#).
- **Diagram style:** [author's DL course, Units 2–5](#) · Transformer schematic matches the current temporal tutorial.
- **Materials:** [practical](#) · [homework](#) · [full references](#).